01 — MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

PAGE / 08

28 Double-Six Dominoes, 2 Wildcards, 12-Frame Tray & Doubles Lid

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

WHAT IS THIS BOX FOR? STOP COUNTING ON YOUR FINGERS.

Fingers are great for holding sandwiches, not for adding 9 + 5. These 28 dominoes and 12-frame tray train your eyes to see numbers in chunks.

WELCOME TO SUBITIZING & 12-FRAME PLAY

Why count one-by-one when your brain can recognize patterns instantly?

- **28 Classic Dominoes: Every pair from [0|0] to [6|6] (168 dots total!)**
- **Doubles Practice Lid: Snap matching doubles (1+1 to 6+6) right on top**
- **The 12-Frame Tray: A raised bridge at 10 shows how numbers spill over**
- Secret Underside: Flip any tile over to spot the StackCalc register icon

Try this right now: Grab any two tiles and race to call out their total!

THE "MAKE-A-TEN" BRIDGE

A raised tactile ridge between the 10th and 11th spots physically shows how 9 + 3 naturally regroups into 10 + 2 = 12 without counting by ones.

WHAT'S IN YOUR BOX (32 PIECES)

- 28 Double-Six Domino Tiles + 2 Bonus Mystery Blank Tiles
- 1x 12-Frame Regrouping Tray with 10-Bridge Tactile Ridge
- 1x 3x1 Storage Case + 1x Snap-Fit Doubles Practice Lid

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

1. **FIND ALL 6 DOUBLES & SNAP ONTO THE LID:**
   Hunt for [1|1], [2|2], [3|3], [4|4], [5|5], and [6|6] and seat them into the Doubles Lid!
2. **SPIN ANY DOMINO 180° ON YOUR DESK:**
   Take [2|5] (= 7) and rotate it half a turn: it becomes [5|2] (= 7)—proving a + b = b + a!
3. **DOUBLE IN 2 TAPS WITH [ENTER][+]:**
   Type 5 [ENTER] [+] 1 [+] [g][COUNT] on your calculator to solve [5|6] = 11!

BOX ICON = CALCULATOR [G][COUNT] SUBITIZING & [G][A:B] RATIO KEYS

The button icon embossed on your box lid (left) is the exact calculator key ([g][COUNT] & [g][a:b])! Build every lesson with your blocks, then press the key.

NEAR-DOUBLES ([5|6] = 11) ↔ BOX [G][COUNT] ICON

STEP A: HANDS-ON WITH DOUBLES & NEAR-DOUBLES — 1:1 CAD SCENE



Double [5|5]	Near-Double [5|6]	Box Icon Key
5 + 5 = 10	10 + 1 = 11	[g][COUNT] Subitize

▲ Hold [5|6] beside [5|5] on the Doubles Lid: left halves match (5=5), right half has 1 extra dot (10+1=11)!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [g][COUNT] & [g][a

NEAR-DOUBLE TREE: [5|6] = [5|5] + 1 = 11

DOMINO [5|6] = 11

Double [5|5] = 10 | + 1 Extra Pip

[ENTER] [+] | 1 [+] | [g][COUNT]=11

5 [ENTER] [+] 1 [+] [g][COUNT] → 11

PRESS BOX ICON & RPN KEYS (SC6):

1. DOUBLE 5: 5 ENTER × = 10
2. NEAR-DBL: 1 + g COUNT = 11

✓ DISPLAY (X) = 11

KEYSTROKES: 5 [ENTER] [+] 1 [+] [g] [COUNT]
In RPN, pressing [ENTER][+] doubles 5 into 10 in two taps—then press 1 [+] [g][COUNT] for 11!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

02 — HOW IT WORKS: THE CORE IDEA

PAGE / 08

Triangular Number Set: T(7) = 28 Tiles (168 Dots Total)

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

TRIANGULAR NUMBERS & COMBINATORICS

Why are there exactly 28 tiles in a double-six domino box?
Because pairing numbers from 0 to n forms a triangular number:

T(n+1) = (n + 1)(n + 2) / 2

For double-six dominoes (n = 6):

T(7) = (7 × 8) / 2 = 28 Unique Tiles

And if you count every single dot across all 28 tiles:

Total Dots = 8 × (0 + 1 + 2 + 3 + 4 + 5 + 6) = 8 × 21 = 168 Dots

Every number from 0 to 6 appears on exactly 8 half-tiles!

THE SECRET BELL CURVE OF DOMINO SUMS

Tile sums run from 0 [0|0] to 12 [6|6]. Notice that 6 is the most common sum of all (appearing 4 times: [0|6], [1|5], [2|4], [3|3]). Stacking tiles by their sum builds a real-life probability histogram right on your table!

WHY DOUBLES UNLOCK NEAR-DOUBLES

Once your brain knows 6+6=12 by heart, tricky neighbors like 6+7 become effortless: just think "6+6 is 12, plus 1 more is 13!" Matching the 6 doubles into the lid sockets builds those mental anchor points automatically.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

- ✗ **TRAP 1 — Counting every dot one-by-one:**
  ✓ FIX: Standard dice/domino pip patterns let your visual cortex recognize 5 + 4 = 9 in 0.2 seconds!
- ✗ **TRAP 2 — Struggling with '7 + 8' or '5 + 6':**
  ✓ FIX: Hold [5|6] next to [5|5]: you immediately see it is just a Double Plus One (10 + 1 = 11).
- ✗ **TRAP 3 — Thinking 4 + 3 and 3 + 4 are two facts:**
  ✓ FIX: Rotating one physical domino tile 180° proves both facts are the exact same object!

CALCULATOR CHECK: 5 [ENTER] [+] 1 [+] [g] [COUNT] → X = 11

DOMINO & TRAY QUICK REFERENCE

PIECE	SIZE	ROLE	COLOR
Domino Tile	39.2 × 19.6 × 7 mm	[a | b]	White
Blank Tile	39.2 × 19.6 × 7 mm	Wildcard	White
12-Frame Tray	126 × 42 × 10 mm	10 + 2	Blue
Doubles Lid	126 × 42 × 10 mm	1+1..6+6	Black
3x1 Storage Case	126 × 42 × 49 mm	3x1 Case	Black

BUILT FOR HANDS-ON TABLETOP PLAY

- Deeply recessed high-contrast dots you can count with your eyes closed.
- Sized to match the universal StackCalc grid (each tile = 2 miniblocks).
- Flip any tile over to use it as a mystery number or RPN stack token.
- Snug snap-on lid keeps all 30 tiles quiet inside a backpack.
- Easy to clean: wipe with a damp cloth or rinse in lukewarm soapy water.

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

SOCKET (1.85mm) ▲ 1.85mm
1U = 7.00 mm
VISIBLE DATUM
7mm
BASE PLUG (1.85mm) ▼ 1.85mm

✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely: Only 7.00mm true heights add!

Box Button Icon Key:
[g][COUNT] Subitizing & [g][a:b] Ratio

Triangular Number Set:
T(7) = 7 × 8 ÷ 2 = 28 Unique Tiles

Doubles Fact Anchors:
[n|n] = 2n ↔ n [ENTER] [+]

Near-Doubles Shortcut:
[n | n+1] = 2n + 1 (Double + 1)

Proportional Pip Ratio:
[.] a [.] b [g][a:b] [g][SIMP]

COMBINATORICS MADE TACTILE • EVERY PAIR FROM [0|0] TO [6|6]

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

03 — EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Doubles Fact Matching on the Lid: [1|1] Through [6|6]

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

1 DOUBLES FACTS MATCHING ON LID

1. POP THE DOUBLES PRACTICE SNAP LID:
The top surface of your storage box lid features 6 pairs of recessed sockets labeled [1+1] through [6+6].

2. HUNT FOR THE 6 DOUBLE TILES:

- Find [1|1] → Snap into socket 1+1 (Sum = 2)
- Find [2|2] → Snap into socket 2+2 (Sum = 4)
- Find [3|3] → Snap into socket 3+3 (Sum = 6)
- Find [4|4] → Snap into socket 4+4 (Sum = 8)
- Find [5|5] → Snap into socket 5+5 (Sum = 10)
- Find [6|6] → Snap into socket 6+6 (Sum = 12)

3. PRACTICE RAPID MENTAL DOUBLING:
Tap each double in sequence. State the sum out loud without counting dots!

WHY DOUBLES ARE THE KEY TO ARITHMETIC FLUENCY

Once a student knows 6 + 6 = 12 automatically, difficult near-doubles like 6 + 7 become trivial mental calculations: (6 + 6) + 1 = 13. The doubles lid physically locks these key anchors into place.

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [g][COUNT] & [g][a

STEP 1 (COMPUTE 7×8):
Press 7 [ENTER] 8 [×] → X = 56 (if every ordered pair [a|b] and [b|a] were separate).

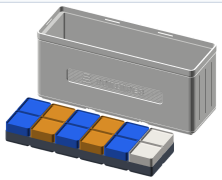
STEP 2 (DIVIDE BY 2):
Press 2 [÷] → X = 28 unique domino tiles (since rotating [a|b] by 180° gives [b|a]!).

STEP 3 (PRESS [g][COUNT]):
Press [g] [COUNT] to inspect the subitized counter pattern on your SC-6 display!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 28 → DISPLAY (X) = 28

LESSON 1: 28 DOUBLE-SIX TILES ↔ [G][COUNT] KEYSTROKES

STEP A: HANDS-ON WITH DOMINOES ON LID — 1:1 CAD SCENE



28 Unique Tiles	6 Doubles Tiles	Tile Footprint
[0|0] Through [6|6]	Sums: 2, 4, 6, 8, 10, 1	39.2 × 19.6 mm (2U×1U)

▲ Every possible pair from [0|0] to [6|6] appears once—snap the 6 Doubles tiles into the lid sockets to anchor mental math!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [g][COUNT] & [g][a

TRIANGULAR COMBINATORICS: T(7) = 28 TILES

28 TILES (168 DOTS)

7 × 8 ÷ 2 = 28 | 6 Doubles [n|n]

[1|1]=2 | [3|3]=6 | [5|5]=10 | [6|6]=12

7 [ENTER] 8 [×] 2 [÷] [g][COUNT] → 28

PRESS BOX ICON & RPN KEYS (SC6):

1. T(7) FORMULA: 7 ENTER 8 × 2 ÷
2. BOX ICON KEY: g COUNT = 28 Tiles

✓ DISPLAY (X) = 28

KEYSTROKES: 7 [ENTER] 8 [×] 2 [÷] [g] [COUNT]
Compute the triangular number T(7) = 7×8÷2 = 28 tiles and press [g][COUNT] to view counters!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 28 → X = 28

ADDITION DOUBLES ANCHORS • SNAP DOUBLES DIRECTLY ONTO THE LID

CONTINUE RIGHT → PAGE 04

04 — EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Bridging Through 10 on the 12-Frame: (9 + 1) + 4 = 14

01 02 03 04

SKU: DOMI-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

2 BRIDGING THROUGH 10 ON THE 12-FRAME

THE PROBLEM: Add domino [5|4] (9) and domino [3|2] (5).

1. SEAT DOMINO [5|4] INTO THE 12-FRAME TRAY:
Its 9 pips (5 + 4 = 9) fill the first 9 sockets of the tray.
Notice: Exactly 1 socket remains before the red 10-Bridge line!

2. REGROUP FROM DOMINO [3|2] (5 PIPS):
- Move 1 pip across to reach the 10-Bridge line (9 + 1 = 10!).
- The remaining 4 pips cross the red ridge into the overflow sockets.

→ (9 + 1) + 4 = 10 + 4 = 14

✓ ASSOCIATIVE PROPERTY OF ADDITION

(A + B) + C = A + (B + C)

9 + 5 = 9 + (1 + 4) = (9 + 1) + 4 = 10 + 4 = 14

The tactile ridge physically forces the decomposition of 5 into 1 + 4. Students develop permanent mental arithmetic bridges instead of **counting by ones on their fingers!**

★ 2-PLAYER GAME: 'SUMS OF 7 SPEED HUNT' (2 MINS)

Spread all 28 dominoes face-up on the table. On 'GO!', both players race to grab every tile whose left and right pips add up to 7 ([0|7] doesn't exist—only [1|6], [2|5], and [3|4]!). **Then verify the total pip sum on your RPN calculator!**

⊙ WHERE THIS SHOWS UP IN COMBINATORICS & NETWORK GRAPH THEORY

Why are there 28 tiles in a double-six set? Because choosing 2 numbers from {0..6} with replacement is the triangular number T(7) = 7×8/2 = 28! Computer scientists use this exact **formula to count handshakes and network connections between servers.**

LESSON 2: BRIDGING 10 ([5|4]+[3|2]=14) ↔ RPN STACK

STEP A: HANDS-ON WITH 12-FRAME TRAY — 1:1 CAD SCENE



Domino [5|4] = 9	Domino [3|2] = 5	Total: 9 + 5
Needs +1 for 10	Splits into 1 + 4	10 + 4 = 14

▲ Place dominoes into the 12-Frame Tray to decompose part-whole sums and see why 9 + 5 regroups into 10 + 4 = 14!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [g][COUNT] & [g][a

PART-WHOLE DECOMPOSITION: [5|4] + [3|2] = 14

[5|4] = 9 | [3|2] = 5

[ENT] 4 [+] | [ENT] 2 [+] | [+] = 14

5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] → 14

PRESS BOX ICON & RPN KEYS (SC6):

1. TILE [5|4]: 5 ENTER 4 + = 9
2. + TILE [3|2]: 3 ENTER 2 + + = 14

✓ DISPLAY (X) = 14

KEYSTROKES: 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+]
Sum [5|4]=9 into Y and [3|2]=5 into X, then press [+] to combine both dominoes into 14!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 14 → X = 14

PART-WHOLE DECOMPOSITION • SEE WHY 9 + 5 BECOMES 10 + 4

STEP 3: FLIP SHEET OVER ↻ PAGE 05

05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Near-Doubles & Sum Histogram + 3 Desk Challenges

05 06 07 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

3 NEAR-DOUBLES: USE [5|5] TO SOLVE [5|6] IN 1 SECOND

THE GOAL: Solve 5 + 6 without counting dots—just compare against a double!

1. SNAP [5|5] INTO THE LID:
Seat domino [5|5] into its 5+5 = 10 socket on the lid.

2. PLACE [5|6] RIGHT BESIDE IT:
Hold domino [5|6] next to [5|5] and compare the two halves.

3. SPOT THE +1 DIFFERENCE:
Left halves match (5 = 5), right half has 1 extra dot: 10 + 1 = 11!

✓ WHY DOUBLES UNLOCK EVERY NEIGHBORING FACT

Kids who memorize doubles (4+4=8, 5+5=10, 6+6=12) get 12 more facts for free!
Whenever two numbers are neighbors (like 6 + 7), it's just a Double Plus One.
Instead of counting 7 fingers past 6, your eyes see [6|6] + 1 = 13 immediately.
Use the 2 blank wildcard tiles with a dry-erase marker to invent your own games!

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands
remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE

NEAR-DOUBLE TREE: [5|6] = [5|5] + 1 = 11

DOMINO [5|6] = 11

Double [5|5] = 10 | + 1 Extra Pip

5 [ENTER] [+] | 1 [+] | g[COUNT]=11

5 [ENTER] [+] 1 [+] [g][COUNT] → 11

STEP B: PRESS BOX ICON & RPN KEYS (SC6)

TACTILE KEYSTROKE WALKTHROUGH:

1. DOUBLE 5: 5 ENTER + = 10

2. NEAR-DBL: 1 + g COUNT = 11

✓ CALCULATOR DISPLAY (X) → 11

FULL KEYSTROKES: 5 [ENTER] [+] 1 [+] [g] [COUNT]

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 11 → X = 11

3 DESK CHALLENGES — TRY THESE NOW!

WARM-UP #1 Match all 6 doubles ([1

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #1 below!

LEVEL-UP #2 Find all 4 dominoes in the box whose dots add up to 6!

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #2 below!

BRAIN-MELTER #3 Place [6

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#3: [1] to [6|6]) on the Tile! | Sums are 2, 4, 6, 8, 10, 12 (All EVEN!) | #2: [0|6], [1|5], [2|4], and [3|3] | #3: 2] (8) and [2|2] (5) on the 12-Frame. What crosses the 10-bridge? | 10 + 3 = 13

FOR PARENTS & TEACHERS (Grades K–4 (Subitizing, Addition Doubles & Mental Math))

- CCSS.MATH.CONTENT.1.OA.C.6: Add and subtract within 20 using doubles and making ten.
- CCSS.MATH.CONTENT.2.OA.B.2: Fluently add and subtract within 20 using mental strategies.
- TEKS 111.3.b.3.D: Apply basic fact strategies to add and subtract within 20.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 BLOCKS → BOX ICON KEY SPEED-RUN!

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([g][COUNT] & [g][a:b]) and keystrokes on your calculator:
   • 5 [ENTER] [+] 1 [+] [g] [COUNT] → X = 11

CHECK OFF YOUR MASTERY MILESTONES:

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

05 06 07 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its
labeled home bay below so everything nests flush and the lid snaps shut:

Col 1 (6x) | Col 2 (6x) | Col 3 (6x) | Col 4 (6x) | Col 5 (6x)

✓ FIVE COLUMNS OF 6 TILES (30 TILES TOTAL)

The storage box holds 5 neat vertical stacks of 6 domino tiles each (5 × 6 = 30 tiles:
all 28 double-six dominoes plus the 2 blank wildcard tiles).
When all 30 tiles are stacked in the 5 columns, every stack sits flush with the
top rim—so a single missing domino is obvious before you snap the lid on!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE

SELF-CHECKING RIM & WATERLINE CROSS-SECTION

✓ FLUSH 84.0 MM = LID CLOSES

✓ OK | ✓ OK | ✓ OK | ✓ OK | ✗ HIGH

5 Columns × 6 Dominoes Flush Rim + 12-

▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall
or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

DOUBLES PRACTICE LID & TRAYNESTING

- The Doubles Lid snaps directly onto the storage box and seats the 12-Frame Tray on top.
- Flip any domino tile over to see the debossed 4-level StackCalc RPN register icon.
- Easy care: wipe clean or rinse in lukewarm soapy water.

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished,
see PAGE 08 (bottom-right) for your complete Part ID Reference Table.

✓ DOMI-01 4-POINT CLEAN-UP CHECKLIST

1. FIVE 6-TILE STACKS: Stack 6 dominoes in each of the 5 columns (28 tiles + 2 wildcards = 30).
2. CHECK RIM LEVEL: Verify all 5 stacks sit completely flush with the box rim.
3. NEST 12-FRAME TRAY: Seat the Blue 6×2 12-Frame Tray and Doubles Practice Lid.
4. TUCK CARD & SNAP: Fold this booklet to Panel [01] and snap the lid closed!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2).
All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero
wasted space, and every embossed box icon matches its calculator operation key!

5 EQUAL COLUMNS OF 6 TILES • SPOT A MISSING DOMINO AT A GLANCE

NEXT: CONNECTED LAB ▾ PAGE 07

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 **Ten-Frames Lab** Early Counting & Regrouping	DOMI-01 ◂ YOUR BOX **Math Domino Studio** Doubles & 12-Frame Part-Whole	FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces
PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers	ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring	TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges
PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner	VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals	SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine

WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact
same height as the 1/12th fraction block and the [2] prime block later on!

1. DOMI-01 BOX LID ICON

Σ+

[g][COUNT] & [g][a:b]

EMBOSSED ON BOX LID

Build First → Press Key!

2. SC-6 ELEMENTARY RPN



3. SC-32 SCIENTIFIC RPN



▲ BOX ICON ↔ CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- Each domino tile (39.2 × 19.6 mm) equals exactly two standard StackCalc unit squares.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Flip any tile over to reveal the 4-register RPN stack icon—linking dot patterns to calculator registers.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides,
and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 28 (T{7} Tiles) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y
2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]
3. PRESS BOX ICON KEY:
Press [g][COUNT] & [g][a:b]
✓ Zero Parentheses • Always Visible!

Σ+

BOX ICON = [g][COUNT] & [g][a:b] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 7 [ENTER] 8 [×] 2 [÷] [g] [COUNT] → X = 28
• Lesson 2: 5 [ENTER] 4 [+] 3 [ENTER] 2 [×] [+] → X = 14

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

08 — KEEP THIS CARD! PART IDs & VIDEO

PAGE / 08

Complete 32-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: DOMI-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

PART ID REFERENCE TABLE (32 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card
in your lid so you always have the exact Part IDs below for single-piece refills!

PART NAME	QTY	COLOR / FINISH	PART ID
Double-Six Domino Tiles	28x	White	DOMI-TILE-01a
Blank Wildcard Dominoes	2x	White	DOMI-BLNK-01a
6x2 12-Frame Tray	1x	Blue	DOMI-TRAY-d4e
Doubles Practice Snap Lid	1x	Black	DOMI-LID-b7c
3x1 Storage Case	1x	Black	DOMI-BOX-a1b

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.

Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

White — Domino Tiles (28x + 2x)
Blue — 12-Frame Regrouping Tray
Black — 3x1 Storage Case
Gray — Doubles Practice Lid

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)

Measure any loose piece in 7mm units (U) against this scale bar to identify it:

0U (0mm) 1U (7mm) 2U (14mm) 3U (21mm) 4U (28mm) 6U (42mm) 9U (63mm) 12U (84mm)

✓ DISTRICT PO & SAFETY: 10-Pack SKU DOMI-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO

stackcalc.io/products/math-dominoes/

See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

Designed in Mountain View, CA • Non-Toxic & Washable

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-6 Guide Ch. 2 (Domino Pairs & Ratios) • SC-32 Manual Ch. 2
STANDARDS: CCSS K.OA.A.3, 1.OA.B.3, 2.OA.B.2, 6.RP.A.1 • TEKS 1.3C, 6.4B (Pairs & Ratios)
WATERLINE SPEC: 5 Columns × 6 Dominoes Flush Rim + 12-Frame Tray

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SCG):
Ex 1: 7 [ENTER] 8 [×] 2 [÷] [g] [COUNT] → X=28
Ex 2: 5 [ENTER] 4 [+] 3 [ENTER] 2 [×] [+] → X=14
Ex 3: 5 [ENTER] [+] 1 [+] [g] [COUNT] → X=11

Σ+

BOX ICON KEY: [g][COUNT] & [g][a:b]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!